JAYANI SAMARAKOON

Undergraduate in Software Engineering

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PROFESSIONAL SUMMARY

Software Engineering undergraduate with practical experience in full-stack web development, development of Android application development, development of REST APIs , database integration, and software development using Git and GitHub. Hands-on experience in building individual and team-based software projects involving frontend development, backend development, CRUD operations, database management, testing, deployment, and SDLC practices. Familiar with Python, Java, Kotlin, JavaScript, React.js, FastAPI, Flask, SQL, Firebase, SQLite, Docker, and Git. Also have experience in AI/ML and MLOps, including machine learning model development, model deployment, and automated ML pipelines. Currently looking forward to gain experience through a Software Engineer Internship .

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Software engineering *2023-Present*
Sri Lanka Technology Campus (SLTC)

G.C.E. Advanced Level *Year of 2022*
K/MW Pinnwala central college
Technology stream: Engineering technology , Science for technology , Information Technology

TECHNICAL SKILLS

Programming : Python, Java, Kotlin, JavaScript, C, C++, SQL
Frontend : React.js, HTML, CSS
Backend : FastAPI, REST APIs
Mobile : Flutter, Android Studio
AI/Machine Learning : Scikit-learn, Pandas,NumPy ,PyTorch
Databases : MySQL, Firebase
UI/UX : Figma, Wireframing, Prototyping
Tools : Git, GitHub, VS Code, Jira ,Docker,MATLAB,Google Colab,Jupyter

PROJECTS

DeepShield: Explainable Multimodal Framework for Trust Assessment and Evidence-Based Fake News Detection
Python/ PyTorch / Transformer /BERT /CLIP / OpenCV /FAISS / FastAPI /React.js *(Final year project)*

- Developing a multimodal AI system using text, image, and metadata for fake news detection.
- Using RAG and Explainable AI approach to provide evidence-based predictions and a Trust Score.
- Building REST APIs using FastAPI and a responsive frontend using React.js.

Mobile Application - MealMate *(Ongoing - Individual)*

Android / Java / XML/ Android Studio/Flutter

- Developed a user-friendly mobile application for browsing and ordering food.
- Developed user authentication with login and register features.
- Designed responsive and intuitive user interfaces to improving user experience.
- **Github :** github.com/-MealMate-

Mobile Application – LankaSmartMart *(Group)*

Figma/Android Studio /Java/Kotlin /MySQL/Git/GitHub

- Developed an Android app that has UI/UX designed by using Figma.
- Connected with Firebase/MySQL for dataaccess and storage .
- Collaborated via Git/GitHub and followed SDLC principles of development, testing, and deployment.
- **Github:** LankaSmartMart---Android

Mobile Application - EduTrack *(Individual)*

Kotlin/ Jetpack Compose / SQLite /Android Studio /Git & GitHub

- Developed an Android student management application using Kotlin and Jetpack Compose.
- Implemented CRUD operations, student search, and SQLite database integration.
- **GitHub:** https:EduTrack_App2

Web application - Inventory Management & Stock Prediction System *(Individual)*

Python/Flask/SQLite/ HTML/CSS/JavaScript

- Developed a full-stack inventory management app using Flask framework.
- Implemented CRUD operations, responsive dashboard, and SQLite database integration.
- **Github:** Inventory-management-system

Web Application - Synthrium *(Individual)*

Python /FastAPI /Roboflow /Computer Vision /HTML /CSS /JavaScript

- Created a web application to detect fruits and vegetables using image upload and live camera .
- Created REST APIs using FastAPI and integrated Roboflow computer vision model.
- Built a responsive interface and configured Vercel deployment.
- **Github** : AI-fruit-vegetable-detection

ADDITIONAL PROJECTS - AI/ML/DL

CardioAI – Heart Attack Risk Prediction System *(Individual)*

Python / Scikit-learn / Streamlit / Pandas

- Built an end-to-end ML prediction system using 1,000 records and 9 cardiovascular features.
- Compared 4 different classification models and reached 86.19% ROC-AUC and 66.67% recall
- Developed a Streamlit dashboard for prediction and model evaluation.
- **Github** : CardioAI-Heart-Attack-Risk-Prediction-

Hospital Appointment No-Show Prediction Using Machine Learning *(Group)*

Python/Pandas/NumPy/Scikit-learn/Matplotlib/Jupyter Notebook /Streamlit/Joblib

Responsibilities:

- Designed an algorithm based on machine learning that could predict patients' absence in appointments based on healthcare data.
- Developed a Streamlit web application that can predict patient no-shows in appointments in real time.
- **Github :** smartcare-ai-project

SMS SPAM Detection system *(Group)*

Python/ Random Forest/ LSTM/Flask/FastAPI /HTML/CSS/JavaScript/ Scikit-learn/TensorFlow

Responsibilities:

- Designed and trained a Random Forest classifier that detects spam messages.
- Designed and optimized the LSTM based deep learning model for text sequential classification.
- Designed and developed a responsive web interface that predicts SMS spam in real time.
- **Github :** NLP-SMS-Spam-Detection

SOFT SKILLS

- **Team Collaboration & Leadership**
  Able to work effectively in teams and able to take initiatives to achieve project objectives.
- **Project Management**
  Plans, organizes, and completes tasks in an efficient manner.
- **Learning Agility**
  Able to learn new tools, technologies, and techniques.

REFERENCES

Available upon request.